Network administration and Monitoring

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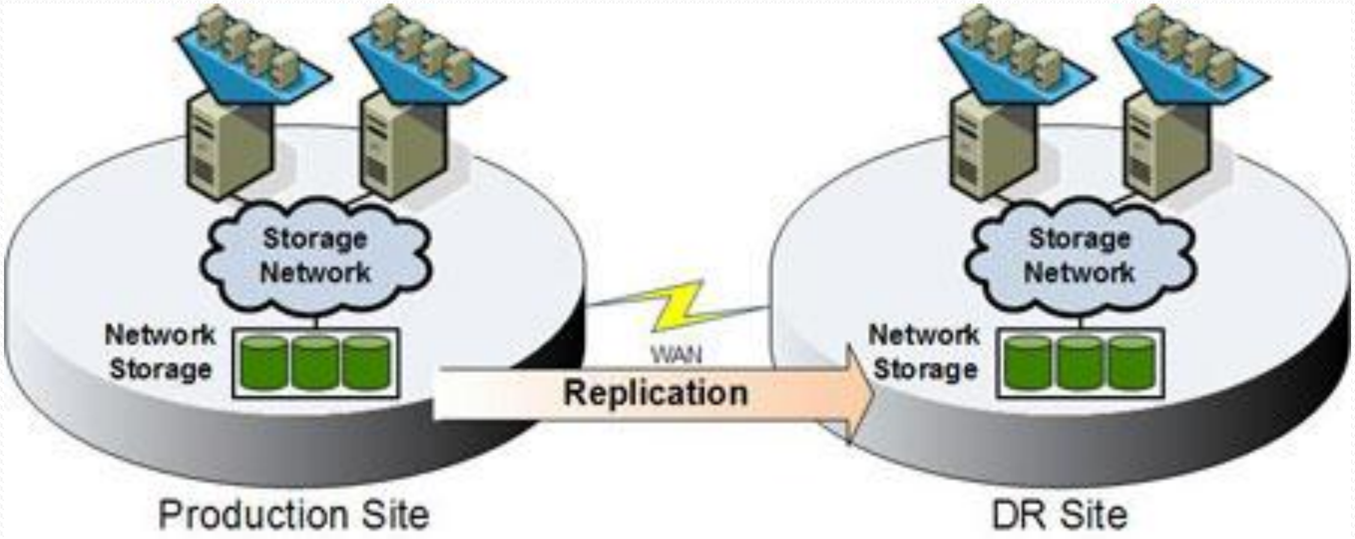
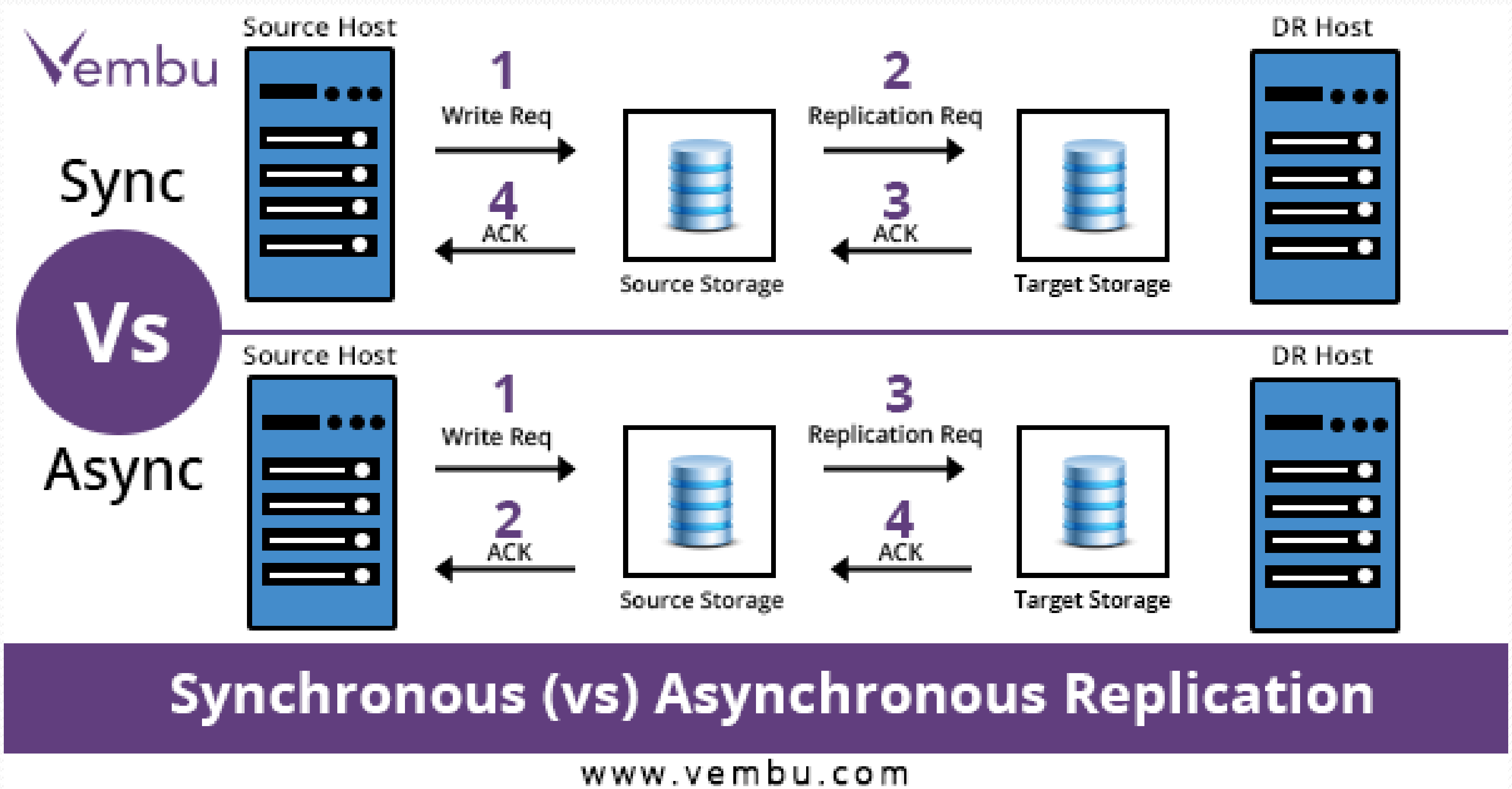
Replication

- What is Replication?
  - ✓ **Data Replication** is the process of storing data (or incremental changed data) in more than one site or node periodically based in RPO & RTO. It is useful in **improving the availability of data**. It is simply copying data from a database from one server to another server so that all the users can share the same data without any inconsistency. The result is a **distributed database** in which users can access data relevant to their tasks without interfering with the work of others.



Replication

- Replication Type:
  - ✓ Sync: RPO=0 (in banking) (Data must be same in both sites) (High cost) (Latency = 5-10)
  - ✓ Async: (Replication every period of time, RPO = constant time (e.g. 40 minute) based in business) (Latency not important) (bandwidth must guaranty transfer data change based in RPO)

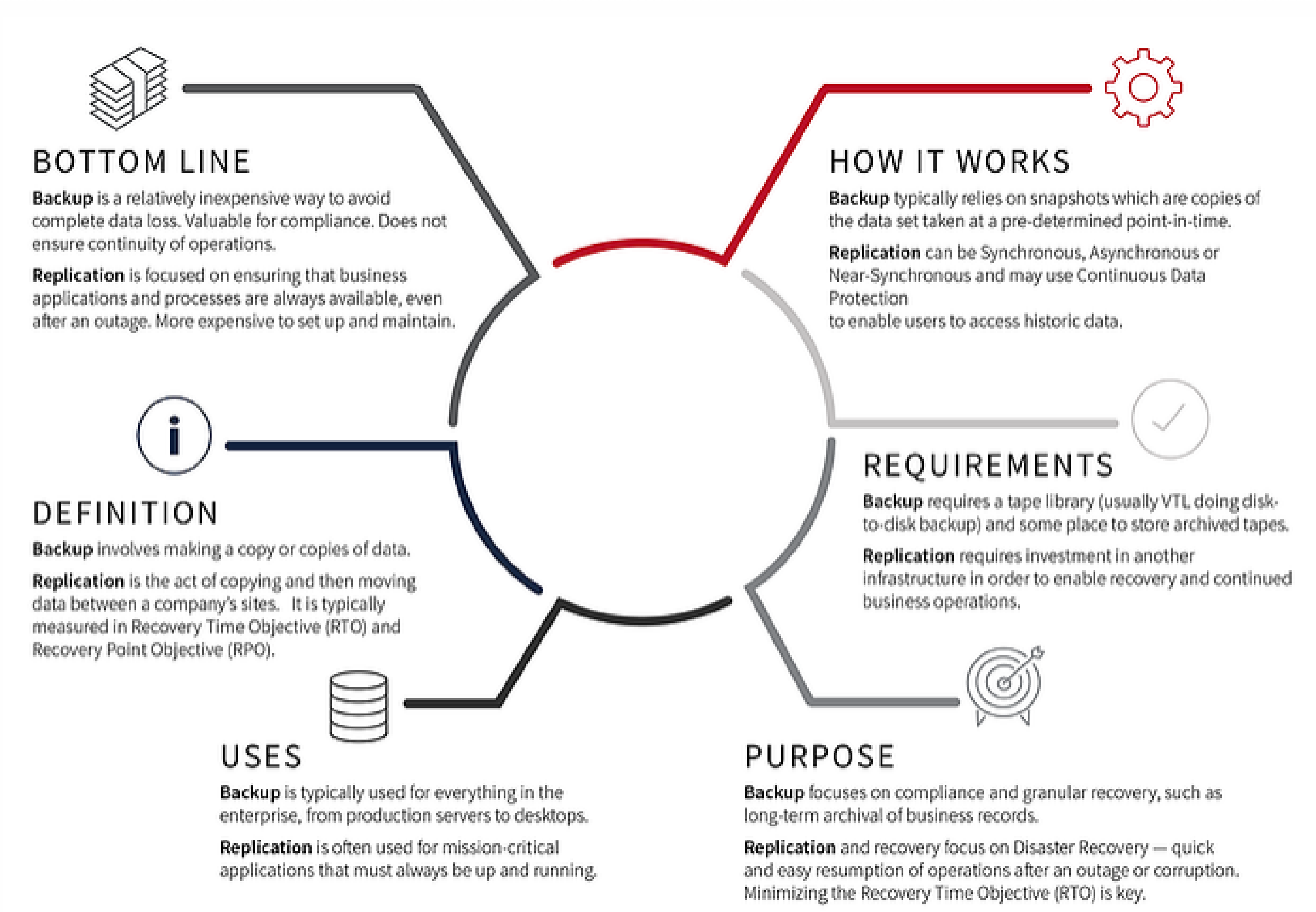


Replication

- Replication Level
  - ✓ Application Rep.
  - ✓ OS Rep.
  - ✓ VM Rep.
  - ✓ Storage (Lun) Rep.
- Replication Location
  - ✓ In same Data center
  - ✓ In different Data center
  - ✓ In remote sites
  - ✓ In clouds

Replication

- Replication vs Backup.

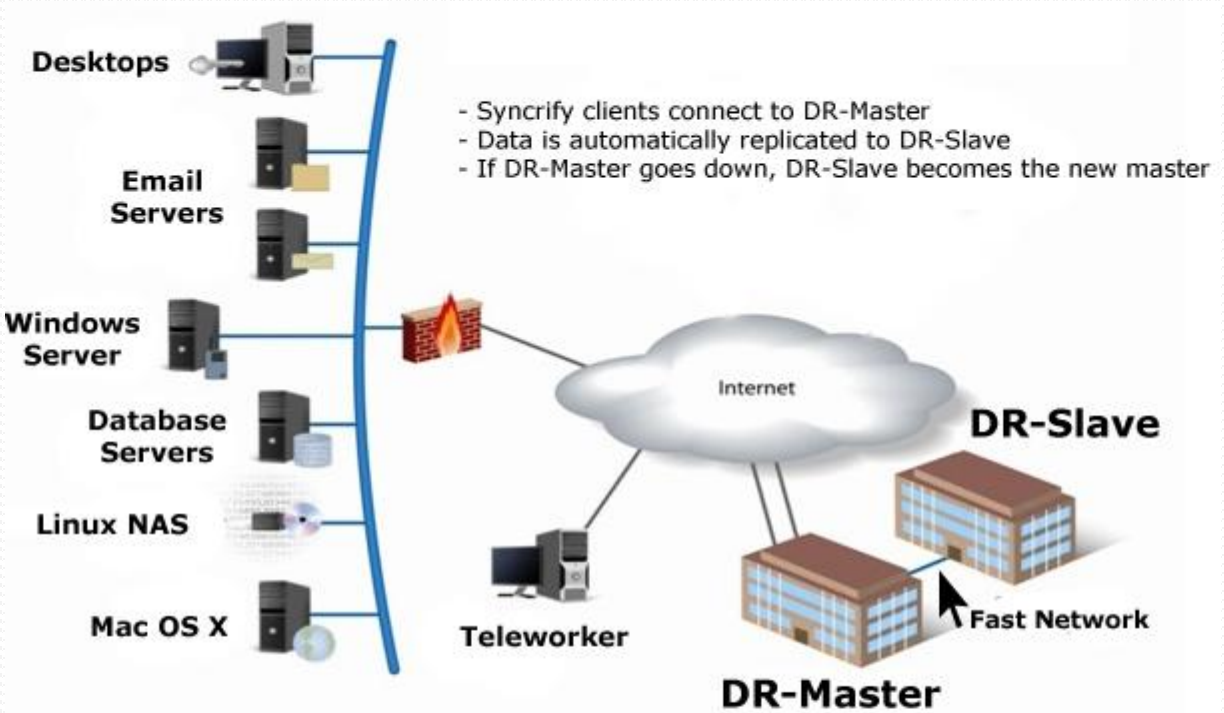


Disaster Recovery (DR)

- Disaster recovery is a **methodology** that describes the **steps that you need to perform once a disaster has occurred**, to bring **data**, **services**, and **servers** back to an **operational state**. An effective disaster recovery plan addresses the organization's needs without providing an unnecessary level of coverage. While absolute protection may seem desirable, it is unlikely to be economically feasible. In creating a disaster recovery plan, **you need to balance the cost to the organization of a particular disaster, with the cost to the organization of protection from that disaster**.

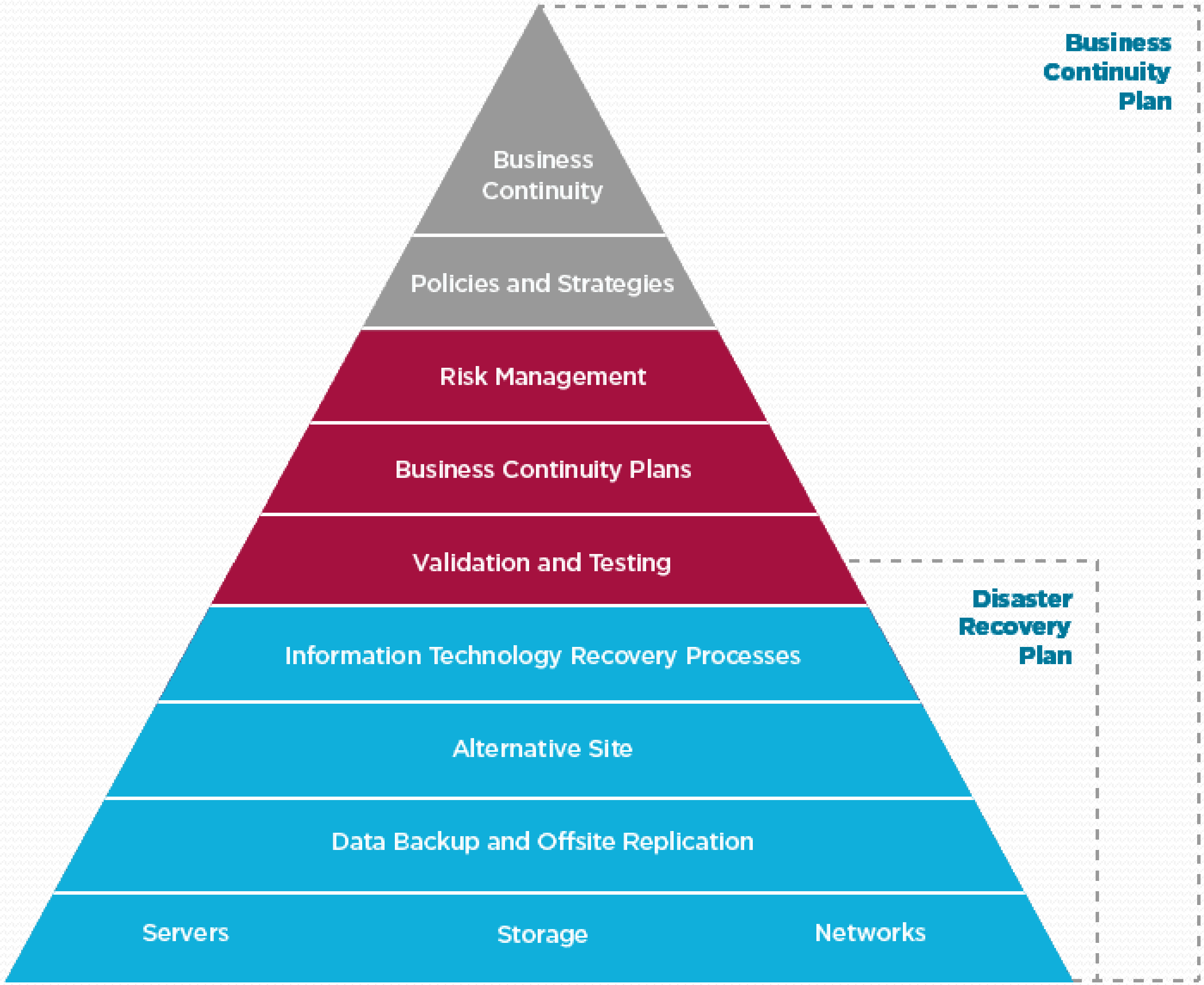
Disaster Recovery (DR)

- In DR-Slave must configure (like DR-Master):
  - ❑ Network
    - ✓ Routing
    - ✓ Switching
    - ✓ Vlan
    - ✓ IP
  - ❑ DNS
  - ❑ NLB
  - ❑ ……



- Must have DR plan & Business Continuity Plan (BCP)

Disaster Recovery (DR)



Example of DR Plan

Disaster Recovery (DR)

- Before developing a disaster recovery strategy, organizations must identify their disaster recovery requirements to ensure that they will provide appropriate protection for critical resources. The following is a high-level list of steps that you can use to identify disaster recovery requirements:

1) **Define organization-critical resources**. These resources include data, services, and the servers upon which the data and services run.
2) **Identify risks associated with those critical resources**. For example, data can be accidentally or intentionally deleted, and a hard drive or storage controller where data is stored might fail. Additionally, services that use critical data might fail for many reasons (such as network problems), and servers might fail because of hardware failures. Major power outages could also cause entire sites to shut down.
3) **Identify the time needed to perform the recovery**. Based on their business requirements, organizations should decide how much time is acceptable for recovering critical resources. Scenarios may vary from minutes to hours, or even a day.
4) **Develop a recovery strategy**. Based on the previous steps, organizations will define a service level agreement that will include information such as service levels and service hours. Organizations should develop a disaster recovery strategy that will help them minimize the risks, and at the same time, recover their critical resources within the minimum time acceptable for their business requirements

Disaster Recovery (DR)

- When planning for backup for your enterprise, you need to develop strategies for recovering data, services, servers, and sites. You also need to make provisions for offsite backup:
  - **Data Recovery Strategies**: Data is the most commonly recovered category in an enterprise environment. This is because it is more likely that users will delete files accidentally, than it is for server hardware to fail or for applications to cause data corruption. Therefore, **in developing an enterprise disaster recovery strategy, take into account small disasters**, **such as data deletion**, **in addition to big disasters, such as server or site failure**. When considering data recovery strategies, backup is not the only technology for data recovery. You can address many file and folder recovery scenarios by implementing previous versions of file functionality on file shares. You can also **replicate data in different physical locations**, or to a **public or private cloud**.
  - **Service Recovery Strategies**: The functionality of the network depends on the availability of certain critical network services. Although well-designed networks build redundancy into core services such as Domain Name System (DNS) and Active Directory® Domain Services (AD DS), even those services might have issues, such as when a major fault is replicated that requires a restore from backup. In addition, an enterprise backup solution must ensure that services such as Dynamic Host Configuration Protocol (DHCP) and Active Directory Certificate Services (AD CS), and important resources such as file shares can be restored in a timely and up-to-date manner.

Disaster Recovery (DR)

- **Full Server Recovery Strategies**: Developing a full server recovery strategy involves determining which servers that you need to be able to recover, and the **RPO** and **RTO** for critical servers. Suppose that you have a site with two computers functioning as domain controllers. When developing your backup strategy, should you aim to have both servers capable of full server recovery with a 15 minute RPO? Alternatively, is it only necessary for one server to be recovered quickly if it fails, given that either server will be able to provide the same network service and ensure business continuity? In developing the full server recovery component of your organization's enterprise backup plan, determine which servers are required to ensure business continuity, and ensure that they are backed up regularly.
- **Site Recovery Strategies**: Most larger organizations have branch office sites. While it might be desirable to back up all the computers at those locations, it may not be economically feasible to do so. Developing a site recovery strategy involves determining which data, services, and servers at a specific site must be recoverable to ensure business continuity.
- **Offsite Backup Strategies**: Many organizations that do not store offsite backups do not recover from a primary site disaster. If your organization's head office site has a fire, is subject to a once-in-a-100-year flood, an earthquake, or a tornado, it will not matter what backups strategies you have in place if all those backups are stored at the location that was destroyed by the disaster.
- A comprehensive enterprise data protection strategy involves moving backed-up

Disaster Recovery (DR)

- A comprehensive enterprise data protection strategy involves moving backed-up data to a safe offsite location so that you can recover it no matter what kind of disaster occurs. This does not need to happen every day. The RPO for recovery at the offsite location—often called the disaster recovery site—is usually different from the RPO at the primary site.

Disaster Recovery (DR)

- When implementing a disaster recovery strategy, **organizations should follow these best practices**:
  - **Perform a risk assessment plan**. This will help you identify all of the risks associated with the availability of your organization data, servers, services, and sites.
  - **Discuss the risks you evaluated with your business managers**. Decide together which resources should be protected with the disaster recovery plan, and which resources should be protected with disaster mitigation, and at which level. The higher the requirements for disaster recovery are, the more expensive they are. You also want to have a low-level disaster recovery plan for resources that are protected with disaster mitigation.
  - **Ensure that each organization has its own disaster recovery plan**.
  - **Document in detail all of the steps that should be performed in a disaster scenario**.
  - **Test your disaster recovery plan on a regular basis in an isolated**, non-production environment.
  - **Evaluate your disaster recovery plan on a regular basis**, and update your disaster recovery plan based on your evaluation.